

2025 AMC 10A Problem 4 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 4. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 4

Problem:

A team of students is going to compete against a team of teachers in a trivia contest. The total number of students and teachers is 15. Ash, a cousin of one of the students, wants to join the contest. If Ash plays with the students, the average age on that team will increase from 12 to 14. If Ash plays with the teachers, the average age on that team will decrease from 55 to 52. How old is Ash?

Answer Choices:

- A. 28
- B. 29
- C. 30
- D. 32
- E. 33

Solution:

Let s = number of students and t = number of teachers. Then

$$s + t = 15$$

We can compute the average age of the students as follows:

$$\frac{12s + A}{s + 1} = 14 \Rightarrow 12s + A = 14s + 14 \Rightarrow A = 2s + 14 \quad (1)$$

and the average age of the teachers:

$$\frac{55t + A}{t + 1} = 52 \Rightarrow 55t + A = 52t + 52 \Rightarrow A = 52 - 3t \quad (2)$$

Since we have 2 values for A in (1) and (2), we can equate them to get

$$2s + 14 = 52 - 3t \Rightarrow 2s + 3t = 38 \quad (3)$$

From $s + t = 15$, $t = 15 - s$. We can substitute this into (3) to get

$$2s + 3(15 - s) = 38 \Rightarrow 2s + 45 - 3s = 38 \Rightarrow -s = -7 \Rightarrow s = 7, t = 15 - s = 8$$

Using (1), we get

$$A = 2s + 14 = 2(7) + 14 = 28$$

which gives us an answer of **(A) 28**.

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